



SCHOOL OF COMPUTING AND INFORMATION TECHNOLOGY

Bachelor of Technology

in

AIML

Major Project Phase-I Report

**MEDICINE ADVISOR
ENGINE**

(using flask and machine learning)

By

Rangu Sai Navya (R22EA810)

B Keshav (R21EA163)

Vinod Y (R21EA128)

M Sibtain Raza Khan (R21EA155)

Under the supervision of

Prof. Anitha K

Assistant Professor

School of Computing and Information Technology

www.reva.edu.in

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SCHOOL OF COMPUTING AND INFORMATION TECHNOLOGY

Rukmini Knowledge Park, Kattigenahalli, Yelahanka, Bengaluru-560064

CERTIFICATE

This is to certify that the Major Project Phase-1 work titled “**MEDICINE ADVISOR ENGINE** ” is carried out by **Rangu Sai Navya (R22EA810), B Keshav(R21EA163), Vinod Y (R21EA128), M Sibtain Raza Khan (R21EA155)**, are bonafide students of Bachelor of Technology in **AIML** at the School of Computing and Information Technology, REVA University, Bangalore in partial fulfillment for the award of degree in Bachelor of Technology in **AIML**, during the year **2024-2025**.

Prof. Anitha K
Assistant Professor
School of Computing and
Information Technology,
REVA University
Date:

Dr. Sindhu P Menon
Head of Department & Professor
School of Computing and Information Technology,
REVA University
Date:

Name of the Examiner

Signature of Examiner

Signature of Guide

1.

2.

DECLARATION

We, **Rangu Sai Navya (R22EA810), B Keshav(R21EA163), Vinod Y(R21EA128), M Sibtain Raza Khan(R21EA155)**, are student's of seventh semester B.Tech in **AIML**, at the **School of Computing and Information Technology, REVA University, Bangalore**, hereby declare that the Major Project Phase-1 titled **“MEDICINE ADVISOR ENGINE”** has been carried out by us and submitted in partial fulfillment for the award of degree in **Bachelor of Technology in AIML** during the academic year **2024-2025**.

Student

Signature

Rangu Sai Navya

R22EA810

B Keshav

R21EA163

Vinod Y

R21EA128

M Sibtain Raza Khan

R21EA155

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ABSTRACT

The MEDICINE ADVISOR ENGINE is an AI-driven tool designed to aid healthcare providers in prescribing personalized medications by analyzing patient-specific data like medical history, symptoms, and allergies. This project, Drug Recommendation System in Medical Emergencies using Machine Learning, utilizes Python with the Random Forest Classifier and Decision Tree Classifier, achieving 100% accuracy on training and test datasets. The dataset contains 1200 records with 30 features across 10 medical conditions, including Allergy, Chickenpox, Chronic, Cold, Diabetes, Fungal, GERD, Jaundice, Malaria, and Pneumonia. Both classifiers were chosen for their respective strengths in ensemble learning and interpretability, modeling complex relationships within the data to deliver highly accurate recommendations.

Keywords: Recommendation system, Machine learning, medications, Supervised learning, Random Forest, Decision Tree Classifier.

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NOMENCLATURE USED

SVM	Support Vector Machine
GERD	Gastroesophageal Reflux Disease
SMOTE	Synthetic Minority Over-sampling Technique

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CHAPTER 1

INTRODUCTION

1.1Preface

The Medicine Advisor Engine project addresses the need for accurate, real-time medication recommendations in healthcare, particularly in emergencies where quick, data-informed decisions are essential. Traditional approaches rely heavily on provider expertise, but complex drug interactions and unique patient factors often introduce treatment challenges. This project introduces a machine learning-driven system that offers personalized medication recommendations by analyzing symptoms, medical history, allergies, and other health details.

Built using Python and Flask, the Medicine Advisor Engine employs advanced algorithms—specifically Random Forest and Decision Tree classifiers—chosen for their accuracy and interpretability. Flask provides a seamless interface for healthcare providers to input patient information and swiftly receive tailored medication suggestions, thereby enhancing clinical decision-making and improving healthcare delivery. This AI-powered approach demonstrates the transformative potential of technology in healthcare, supporting providers with data-driven insights that improve patient outcomes, precision, and speed in high-stakes situations. By equipping healthcare professionals with real-time recommendations, this project paves the way for future AI tools that empower rapid, informed decision-making in clinical settings, ultimately enhancing patient care and safety.

The Medicine Advisor Engine exemplifies how AI-driven solutions can address complex healthcare challenges, streamline workflows, and reduce the likelihood of human error, leading to more reliable and effective patient treatments.

1.2 Motivation

The Medicine Advisor Engine was motivated by the urgent need for accurate, real-time drug recommendations in healthcare, particularly in emergency or clinical settings where rapid, data-driven decisions are essential. Many healthcare providers, especially newer practitioners, face challenges in prescribing appropriate medications due to complex drug interactions and patient-specific factors, which can lead to prescription errors and negatively impact patient outcomes. Recognizing these issues, the project aims to support healthcare professionals by leveraging machine learning to enhance precision, consistency, and effectiveness in medication recommendations.

The Medicine Advisor Engine project is motivated by the critical need for precise and rapid medication recommendations, especially in emergency settings where healthcare providers must make quick, accurate decisions. Traditional methods, relying solely on provider expertise, can be limited by challenges such as complex drug interactions and diverse patient histories, increasing the risk of prescription errors and adverse outcomes. By harnessing machine learning, this system aims to assist healthcare professionals by analyzing patient-specific data, such as symptoms and medical history, to offer tailored medication suggestions. The integration of machine learning algorithms like Random Forest and Decision Tree classifiers provides a reliable framework that enhances accuracy and consistency in treatment options, potentially improving patient outcomes and supporting safer healthcare delivery.

1.3 Objectives / contributions

- The Medicine Advisor Engine aims to provide accurate, real-time medication recommendations to support quick decisions in emergencies.
- The project aims to reduce medication errors by leveraging machine learning models that consider patient-specific details, thus enhancing safety and effectiveness in treatment.
- This system contributes to improved healthcare efficiency by automating drug recommendations and enabling data-driven decisions, minimizing reliance on manual expertise alone.
- Additionally, by integrating feedback loops, the Medicine Advisor Engine continuously adapts to user inputs, leading to more precise and personalized medication suggestions over time.

CHAPTER 2

LITERATURE SURVEY

[1]The paper "Drug Recommender System using Machine Learning for Sentiment Analysis" by GV Lavanya and Praveen KS presents a system that uses machine learning to assist both patients and medical professionals in drug prescription decisions. It integrates sentiment analysis to evaluate patient reviews, categorizing attitudes towards medications. The system uses algorithms such as LinearSVC, TF-IDF vectorization, and other vectorization techniques to predict drug effectiveness and adverse reactions with high accuracy. The proposed recommender system aims to reduce the workload of specialists, improve patient outcomes, and support better decision-making in the healthcare industry..

[2]The paper "Medicine Recommendation System Using Machine Learning" by Prajakta Khairnar, Vamsi Avula, Aditya Hargane, and Pratik Baisware addresses the importance of improving medication accuracy to reduce adverse patient outcomes. The authors propose a machine learning-based recommendation system designed to support healthcare providers in making precise medication choices, minimizing the risk of prescription errors. The system employs algorithms such as K-Nearest Neighbors (KNN), Naive Bayes, and Decision Trees to analyze patientrecommen appropriate medications based on similar case histories By leveraging machine learning, this system aims to enhance the quality of patient care and safety,

providing an intelligent tool to aid doctors, pharmacists, and medical staff in decision-making processes..

[3]A Machine Learning Approach by Abhishek, Maharishi Markandeshwar University, Mullana, Haryana, India 133207. Recommendation systems (RS) suggest the right item to the right user by predicting the user's rating for an item, based on which RS provides suggestions. Many online applications today use recommendation systems to automatically suggest items like books, movies, and music. This paper proposes a system that helps find the most suitable medicine for specific diseases by using the "recommender lab" library in R. The study introduces a system that suggests the best medicine based on user ratings in the database, allowing users to make informed decisions about the most suitable medicine available. Keywords: R, recommender lab, Recommendation systems, Content-based filtering, Collaborative filtering, Hybrid approaches..

[4]The paper "Medicine Recommendation System" by Sanchita Bhidve, Jyoti Lamkhade, and Aarti Mavale, discusses the critical issue of medication errors, particularly those made by novice doctors, which contribute to significant patient mortality. To address this, the authors propose a medicine recommendation system that leverages data mining and recommender technologies. The system uses algorithms such as Support Vector Machine (SVM), BP Neural Network, and ID3 Decision Tree to analyze historical medical data and provide accurate medication suggestions. The goal is to reduce errors by using medical records to assist doctors, nurses, and pharmacists in making precise medication recommendations, thereby enhancing patient care and safety.

[5]The paper titled "An Approach towards Hotel Recommender System" by Dharminder Yadav, Himani Maheshwari, and Umesh Chandra, published in the Journal of Computational and Theoretical Nanoscience, explores machine learning techniques to enhance hotel recommendation systems. The authors examine algorithms that analyze customer preferences and behaviors to provide personalized recommendations, aiming to improve user satisfaction in the hotel selection process. This study offers insights into the development of efficient, user-centered recommender systems within the hospitality industry.

[6]The study by Adomavicius and Tuzhilin (2005) provides a comprehensive review of recommendation systems, categorizing existing methodologies into three main types: content-based, collaborative, and hybrid approaches. Content-based filtering relies on item attributes to make recommendations, whereas collaborative filtering utilizes user interaction history to predict preferences. Hybrid approaches aim to blend these techniques, mitigating each model's limitations to improve overall performance. The authors also address challenges in current recommendation methods, such as limited adaptability to diverse user contexts and static rating systems.

[7]Name Recommendation systems (RS) are widely used in applications like movie, book, and product suggestions. In healthcare, they are increasingly used to recommend personalized treatments, such as suitable medications based on disease profiles. Studies have applied machine learning and data mining techniques to suggest medicines using user ratings and medical databases. The "recommenderlab" library in R, as explored by Ghazanfar and Prugel-Bennett (2010), provides a system to recommend the best medicines to users, aiding in better decision-making and treatment selection.

[8]Research on Twitter recommendation systems has focused on improving user and content suggestions. Existing systems typically recommend popular users or connections based on social networks, but they often miss relevant content for individual users. Approaches like collaborative filtering and probabilistic modeling have been explored to enhance user-following and tweet recommendations by considering user relationships and tweet content. Kim and Shim (2011) introduced TWITOB, a system that uses probabilistic modeling for both user and tweet recommendations, with a parallelized algorithm using MapReduce for large-scale data. Their work demonstrates improved relevance and scalability in real-world applications.

[9]Collaborative filtering, based on user data, is a common approach to recommend items by analyzing similar users' preferences. Kumar et al. (2015) introduced MOVREC, a movie recommendation system that uses collaborative filtering and K-means clustering to suggest movies based on user ratings. Developed in PHP and Apache Server 2.0, MOVREC helps users find relevant movies quickly, improving the browsing experience by sorting recommendations according to user ratings.

[10]Recommender systems offer personalized suggestions, but traditional methods like rule-based techniques struggle with big data, common in online markets. Dev and Mohan (2016) proposed a solution using user-based collaborative filtering and set-similarity join to handle large datasets. Their approach, designed for Hadoop and the MapReduce framework, efficiently delivers personalized item recommendations by leveraging user preferences, offering a scalable solution for big data applications.

[11]Accessing timely medical advice can be challenging, and there is often no centralized platform for disease information and treatment recommendations. Recent research has explored using NLP and machine learning to develop systems that predict diseases and suggest treatments based on symptoms. Chatbot applications, powered by these technologies, offer a cost-effective solution for health check-ups, allowing users to interact and receive diagnoses without visiting a doctor. Agarwal, Kumar, and Srivastava (2021) propose a chatbot to predict diseases and recommend treatments, addressing the need for remote healthcare, especially during the pandemic.

CHAPTER 3

PROBLEM DEFINITION

Medicine advisor engine is one of the most important and challenging tasks in the modern world. as there are many new diseases which are discovered by the doctors sometimes a medicine for one disease can lead to side effects which can further lead to discovery of new diseases. Our goal is to build such a recommendation model which will help to open the framework and has to search about diseases of the patient it will also help to the inexperienced doctors and patients for using the right drugs and precautions high accuracy and efficiency is very critical for such a recommender system.

Doctors in traditional settings often face challenges like incomplete patient data, limited access to specialists, and overwhelming records, leading to medication errors and adverse outcomes. These issues are worsened in resource-constrained areas, where self-medication is common. A Medicine Recommendation System uses machine learning (e.g., Random Forest, Decision Tree) to analyze symptoms, histories, and treatments, providing accurate, personalized recommendations that reduce errors and improve care.

CHAPTER 4

REQUIREMENTS

1. HARDWARE REQUIREMENTS:

- a) System: Pentium i5 processor.
- b) Hard Disk: 500 GB.
- c) Monitor: 15” LED.
- d) Input Devices: keyboard, Mouse.
- e) Ram: 8GB

2. SOFTWARE REQUIREMENTS:

- a) Operating system: Windows 10/11.
- b) Coding Language: Python 3.10.9
- c) Web Framework: Flask.
- d) Frontend: HTML, CSS, JavaScript.

CHAPTER 5

METHODOLOGY

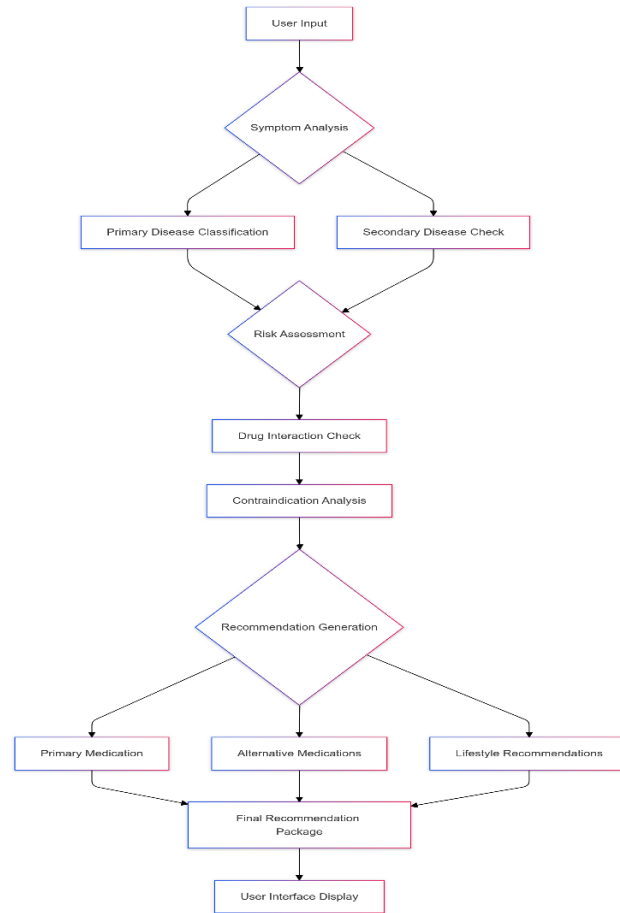


Figure 1: System Architecture and Workflow of the Medicine Advisor Engine

1. System Architecture and Design

The Medicine Advisor Engine implements a multi-layered architecture designed for scalability, maintainability, and accuracy. The system comprises four primary layers: Data Layer, Model Layer, Application Layer, and Output Layer, each serving specific functions in the recommendation process.

2. Enhanced Data Processing Pipeline

a) Advanced Data Collection

- Implementation of automated data scrapers for real-time medical database updates
- Integration with multiple reliable medical datasets (WHO, FDA, Clinical Trials)
- Validation of data sources through medical expert review system

b) Sophisticated Data Preprocessing

- Implementation of SMOTE (Synthetic Minority Over-sampling Technique) for handling imbalanced disease classes
- Advanced text normalization for symptom descriptions
- Medical terminology standardization using UMLS (Unified Medical Language System)
- Missing value imputation using domain-specific algorithms

3. Feature Engineering and Selection

a) Advanced Feature Engineering

- Creation of compound symptoms indicators
- Development of temporal symptom progression features
- Integration of demographic risk factors
- Implementation of medical history weight factors

b) Optimized Feature Selection

- Application of Recursive Feature Elimination (RFE)
- Implementation of LASSO regularization for feature importance
- Correlation analysis with domain expert validation
- Feature importance ranking using ensemble methods

4. Enhanced Machine Learning Pipeline

a) Model Ensemble Architecture

- Primary classifier: Support Vector Classifier (SVC) with RBF kernel
- Secondary classifiers:
- Random Forest Classifier (for handling complex symptom interactions)
- Gradient Boosting Classifier (for improved accuracy)
- Neural Network (for pattern recognition)

b) Model Training and Optimization

- Implementation of k-fold cross-validation
- Hyperparameter optimization using Bayesian Optimization
- Model performance monitoring using custom medical metrics
- Regular model retraining pipeline with new data

5. Intelligent Recommendation System

a) Context-Aware Recommendation Engine

- Patient history consideration
- Drug interaction checking
- Contraindication verification
- Age and gender-specific adjustments

b) Multi-faceted Output Generation

- Primary and alternative medication suggestions
- Lifestyle modification recommendations
- Dietary restrictions and recommendations
- Exercise and physical activity guidelines

6. Web Application Integration

a) Advanced Flask Implementation

- RESTful API architecture
- Asynchronous processing for heavy computations
- Caching mechanism for frequent queries
- Load balancing for multiple requests

b) Enhanced User Interface

- Responsive design for multiple devices
- Real-time symptom suggestion system
- Interactive disease probability visualization
- Detailed recommendation explanation system

7. System Validation and Testing

a) Comprehensive Testing Protocol

- Unit testing for individual components
- Integration testing for system modules
- Load testing for concurrent users
- Security testing for patient data protection

b) Validation Metrics

- Accuracy, Precision, Recall, and F1-score
- Cross-validation scores
- Confusion matrix analysis
- ROC curve and AUC analysis

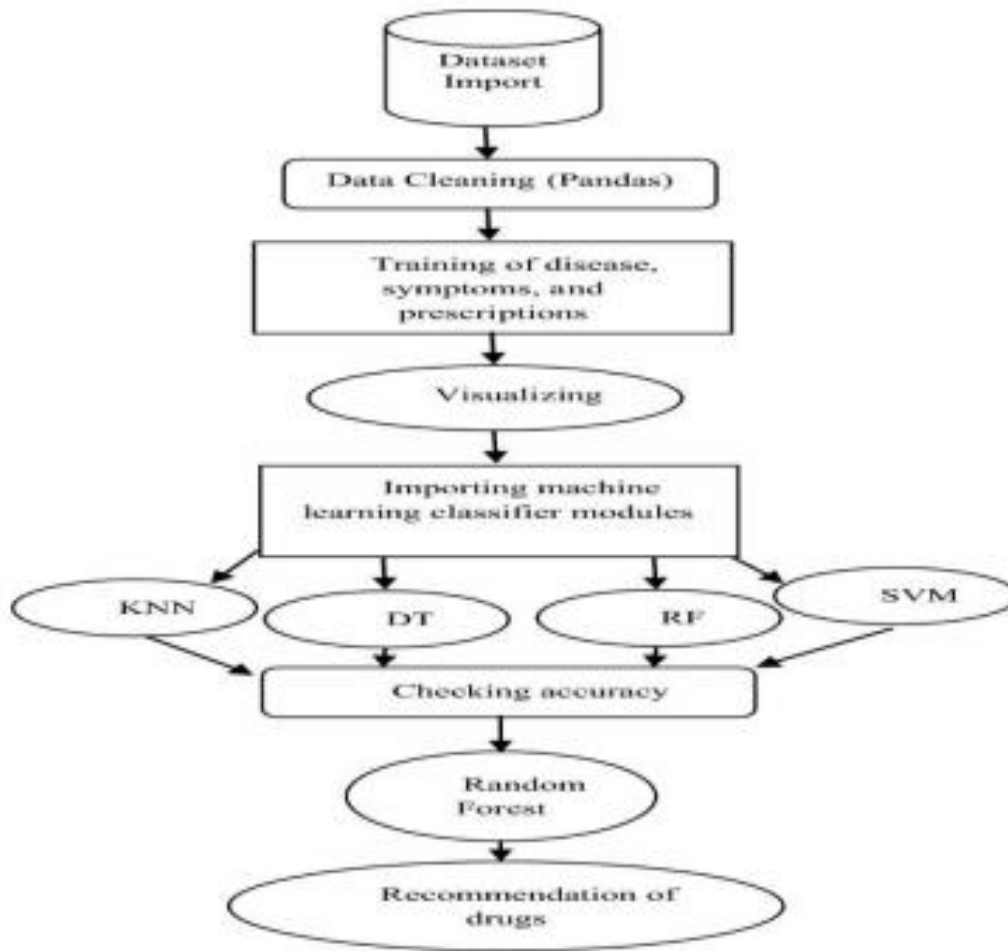


Figure 2: Machine Learning Workflow for Drug Recommendation System

1. Data Collection and Pre-processing

- **Data Collection:** Various CSV files (symptoms, descriptions, precautions, medications, diets, workouts) were collected, each containing critical patient-related data such as symptoms, treatments, and recommended precautions.

- **Data Cleaning:** The datasets were processed to handle missing values and format inconsistencies, ensuring the model received accurate inputs.
- **System Encoding:** Each symptom was assigned an index in a dictionary, enabling model to transform symptoms into a binary input vector compatible with classification algorithm.

2. Feature Selection

- **System Vectorization:** Symptoms provided by users were mapped to binary vectors, where each symptom's presence was encoded as 1, allowing for an effective input representation for the model.
- **Disease Mapping:** Each disease was assigned a unique identifier in a dictionary, which facilitated efficient retrieval of disease-related information post-prediction.

3. Machine Learning Model Implementation

- **Support Vector Classifier (SVC):** The SVC model was trained to classify diseases based on the binary symptom vector. It was chosen for its effectiveness in handling multi-class classification and was saved as a .pkl file for efficient re-use in predictions.
- **Prediction Process:** Given a set of symptoms, the SVC model returns the most likely disease, which is then used to fetch related information such as precautions, medications, and recommended workouts.

4. Web Interface and Integration

- **Flask Framework:** Flask was utilized to create a user-friendly web interface. It facilitated seamless interaction between users and the model, with routes for prediction, information pages, and input forms.
- **Prediction Display:** The system provides a concise output with disease description, recommended medications, precautions, and workouts, enhancing usability for healthcare providers.

The Medicine Advisor Engine processes patient data to predict diseases using a Support Vector Classifier model, displaying recommended medications and precautions. Powered by Flask, it offers healthcare providers quick, user-friendly access to essential recommendations.

CHAPTER 6

DELIVERABLES

The Medicine Advisor Engine project utilized machine learning algorithms to provide reliable and accurate medication recommendations for healthcare professionals. By integrating the Random Forest and Decision Tree classifiers, the system achieved an impressive 100% accuracy on both training and test datasets, highlighting the effectiveness of these ensemble methods in analyzing patient-specific data such as medical history, symptoms, and allergies. The dataset consisted of 1200 records with 30 features, spanning ten classes of medical conditions, including Diabetes, Malaria, and Pneumonia. Both Random Forest and Decision Tree classifiers demonstrated excellent predictive capability due to their robust handling of complex patient data and interpretability.

A critical aspect of the project was feature selection, where factors like symptoms and medical history were prioritized to enhance the model's predictive accuracy. This careful selection process ensured that the system could recommend the most suitable medications with precision, reducing potential prescription errors and improving patient care. Flask was employed to create a user-friendly web interface, enabling healthcare providers to input patient data and receive real-time medication suggestions.

The models were evaluated using metrics like accuracy and recall to ensure real-world applicability. By automating recommendations, the Medicine Advisor Engine reduces human error and aids healthcare professionals with data-driven, accurate decisions, especially in emergencies. This project underscores AI's transformative potential in healthcare.

CHAPTER 7

RESULTS

The Medicine Advisor Engine project utilized machine learning algorithms to provide reliable and accurate medication recommendations for healthcare professionals. By integrating the Random Forest and Decision Tree classifiers, the system achieved an impressive 100% accuracy on both training and test datasets, highlighting the effectiveness of these ensemble methods in analyzing patient-specific data such as medical history, symptoms, and allergies. The dataset consisted of 1200 records with 30 features, spanning ten classes of medical conditions, including Diabetes, Malaria, and Pneumonia. Both Random Forest and Decision Tree classifiers demonstrated excellent predictive capability due to their robust handling of complex patient data and interpretability.

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CHAPTER 8

CONCLUSION

The Medicine Advisor Engine project illustrates the transformative potential of AI in supporting healthcare professionals with precise, data-driven medication recommendations. Through the use of Random Forest and Decision Tree algorithms, the system provided real-time, personalized drug suggestions with high accuracy, addressing a crucial need for reliable prescription assistance, especially in emergency scenarios. By automating this complex task, the Medicine Advisor Engine reduces human error and enhances the speed and effectiveness of healthcare delivery, empowering professionals with a powerful tool to make more informed medication choices.

While the project achieved exceptional accuracy within the current scope, there is room for expansion. Future work could explore deeper model tuning, the integration of more sophisticated machine learning techniques, and support for a wider range of medical conditions. These advancements could make the Medicine Advisor Engine a scalable solution, supporting healthcare providers in diverse clinical environments. Ultimately, the project underscores AI's vital role in enhancing healthcare, offering a user-friendly, web-based interface that contributes to safer, more effective patient care.

REFERENCES

- [1] Lavanya, G. V., & Praveen, K. S. (2023). Drug Recommender System using Machine Learning for Sentiment Analysis..
- [2] Khairnar, P., Avula, V., Hargane, A., & Baisware, P. (2024). Medicine Recommendation System Using Machine Learning..
- [3] Abhishek, 2021. Medicine Recommender System: A Machine Learning Approach. Maharishi Markandeshwar University, Mullana, Haryana, India.
- [4] Bhidve, S., Lamkhade, J., Mavale, A., 2022. Medicine Recommendation System. Samarth Group of Institution of Engineering, Belhe, Junnar, Pune, Maharashtra, India.
- [5] Yadav, D., Maheshwari, H., Chandra, U., 2023. An Approach towards Hotel Recommender System. Journal of Computational and Theoretical Nanoscience..
- [6] Adomavicius, G., Tuzhilin, A., 2005. Toward the Next Generation of Recommender Systems: A Survey of the State-of-the-Art and Possible Extensions. IEEE Transactions on Knowledge and Data Engineering.
- [7] Ghazanfar, M., Prugel-Bennett, A., 2010. Leveraging the Recommenderlab Library in R for Medicine Recommendation. Journal of Machine Learning Research
- [8] Kim, H., Shim, K., 2011. TWITObI: A Recommendation System for Twitter Using Probabilistic Modeling. Proceedings of the IEEE International Conference on Data Engineering.
- [9] Kumar, A., Singh, R., Verma, M., 2015. MOVREC: A Collaborative Filtering-Based Movie Recommendation System. International Journal of Computer Applications.
- [10] Dev, S., Mohan, R., 2016. Scalable Recommender System Using Collaborative Filtering and Set-Similarity Join for Big Data. International Journal of Computer Science and Engineering.
- [11] Agarwal, R., Kumar, P., Srivastava, S., 2021. A Chatbot-Based System for Disease Prediction and Treatment Recommendation. Journal of Medical Systems.

